BAEP 557: Technology Commercialization, Spring 2023
Outside Project Overview and Collaboration

Project Overview

We're excited about our collaboration with the USC Alfred Mann Institute of Biomedical Engineering as well as Ryan Antenucci, a USC IYA graduate and founder of Coralite. Below are some preliminary details with more to come as we continue to meet and work with our outside project teams.

The application area for our semester-long projects will be medical devices and cleantech. We will introduce and explain both sets of technologies, each of which will provide two commercialization options for a total of four project options you'll be asked to rank.

Commercialization Steps

Commercializing a technology (medical, digital, or otherwise) includes a considerable number of risks and challenges. To manage those risks and increase the chances of success, experienced tech transfer professionals like the team at AMI follow a specific, proven set of steps or stages.

Our course builds on those steps and provides a structured but innovative process for technology commercialization supplemented by relevant and validated technology frameworks through course readings and discussions. Together, this process and frameworks help define the commercialization process we will be learning and applying in our course and through our collaborative projects this semester.

Project Stages

The 5 stages in the commercialization process that will define our commercialization projects are (focus areas marked with an *):

1. Selecting and managing the project team
2. Becoming familiar with the technical and market context*
3. Conducting customer discovery research (primary and secondary)*
4. Determining revenue model and pricing, identifying distribution channels (preliminary)
5. Presenting findings to a panel of capital and funding sources (our ultimate objective)

(1) As we launch the projects, we will ask you to: tell us about your skills and expertise; identify your school and degree program; and rank your project choices. Several teams (of 4 to 5 students each) will be allocated across a small handful of projects – two technologies and four opportunities in total.

(2) We'll ask each of you to fill out an NDA and provide each team with relevant primary and secondary research, both technical and market. You'll need to read and familiarize yourself with the material and use it to put together a brief project proposal for the experts and me to approve. (See separate proposal document for requirements.)

(3) You should all be comfortable conducting primary research — in other words, finding and interviewing potential users of the proposed technology to determine if there is an actual need, what benefits the targeted customer is seeking, and what potential product solutions would fit that need and at what price point.

You will be also conducting some secondary research to find the total addressable market (TAM) and come up with a realistic plan on what portion of the market can be obtained (the SOM, Serviceable Obtainable Market) and at what cost. We'll also meet with the inventors or domain experts to walk us through the technology inventions we're looking to commercialize. We may also bring in other experts and advisors throughout the semester to help us get up to speed, for this and the other steps in the commercialization process.

(4) We will gather preliminary data that will helps us design and refine a revenue model. We'll provide you with a basic model. That revenue model will depend on identifying the proper end-users and decision-makers (#4), determining what users are likely to pay, and optimizing distribution channels.

(5) Our ultimate objective is determining whether we can obtain the funding necessary to commercialize a large-scale technology in healthcare and cleantech. Therefore, we will emphasize identifying and attracting funding sources and raising capital throughout our project — with a laser focus on generating the robust data and the resonant 'story' around the technology for potential funders to take an interest and make an investment in a technology-based product or service.

It's also possible that the application you focus on has no ready market, which is useful information and a perfectly acceptable outcome for the projects.

Project Choices and Team Selection

You will have four project choices, based on our two medical devices:

A. Organic nanoparticles (biofilm target), Option 1: *dental* application – 2-3 teams
B. Organic nanoparticles (biofilm target), Option 2: *industry* application – 2-3 teams
C. Environmentally friendly building material, Option 1: *consumer* application – 2-3 teams
D. Environmentally friendly building material, Option 1: *industry* application – 2-3 teams

We'll place you into teams based on your skills, background, and (as far as possible) your top project choices, with an eye toward creating balanced, effective project teams.

Our Expectations of You and Next Steps

We expect you to engage with the course material and learn and apply the process we'll be learning. We also expect you to collaborative effectively with your project teammates and outside project leads and experts. Finally, we expect you to keep all the data we provide and that you discover confidential and private.